Autothrottle, FADEC & Systems Operating Manual — Cirrus Vision SF50 G2 (X-Plane 12)

Release Version: v1.1.0

This operational guide provides real-world and simulated operating procedures for the **Perspective Touch+ digital Autothrottle / FADEC system**, the integrated Garmin GFC 700 digital autopilot, and the **3D Cockpit Electronic Flight Bag (EFB)** on the **Cirrus Vision SF50 G2** in X-Plane 12.

1. Overview of Touchscreen Controls (Center GTC)

On the primary Autothrottle touchscreen page on the Garmin Touchscreen Controller (GTC), there are **8 interactive touch buttons** and **1 dual-concentric rotary knob (**SPEED SET**)**:

Left Column (Speed Management & Master Autothrottle)

- Auto T **(**ON **/** OFF**)**: Master Autothrottle Switch.
  - When **engaged (Green bar)**: The autothrottle actively servos the thrust lever to capture and maintain the selected speed target (KTS or Mach), while automatically enforcing the continuous rating limit (MCT).
  - When **pressed again (Off / Gray bar)**: Instantly disconnects the entire autothrottle system (Auto T, MCT, TOGA, and FMS links), returning **100% manual throttle authority** to your physical hardware throttle lever with no residual servo bias.
- KTS: Selects Indicated Airspeed hold mode in **Knots (**IAS**, range** 85 – 250 KIAS**)**. If switching from Mach mode, it automatically converts the current flight Mach number into the equivalent Indicated Airspeed.
- Mach: Selects speed hold mode in **Mach Number (**0.250 – 0.535 M**)**. If switching from KTS mode, it seamlessly calculates and displays the equivalent Mach number at the current pressure altitude and outside air temperature.
- FMS: Toggles coupling to the FMS automated speed schedule (coupling into the M CRS / E CRS cruise profiles). Pressing it while active transitions back to manual speed selection (MAN) while maintaining the current target speed.
- SPEED SET **Rotary Knob (Bottom-right corner of the touchscreen bezel)**:
  - In KTS **mode**: Adjusts target airspeed in increments of 1 kt (inner knob) or 10 kts (outer knob).
  - In Mach **mode**: Adjusts target Mach in increments of 0.001 M (inner knob) or 0.010 M (outer knob).
  - If the knob is rotated while an automated FMS preset (M CRS or E CRS) is active, the system automatically transitions from FMS mode to manual Mach mode at the newly dialed value.

Right Column (FADEC Thrust Ratings & G2 Cruise Presets)

- TOGA **(Takeoff / Go-Around)**:
  - Smoothly commands the Williams International FJ33-5A turbofan to **100% Takeoff Rated Thrust (**TLA = 1.00**)**, activates the Flight Director to +8° pitch-up target attitude (if off), and pre-sets the climb airspeed bug to 160 KIAS.
- MCT **(Maximum Continuous Thrust)**:
  - Enforces the FADEC **Maximum Continuous Thrust limit (**95% **thrust /** TLA = 0.90**)** and activates the Autothrottle (Auto T ON):
    - If climbing in FLC **(Flight Level Change)** autopilot mode: Holds the throttle locked precisely on the MCT **detent** while autopilot pitch commands maintain the commanded climb airspeed.
    - In **level cruise (**ALT**)**, VS **(Vertical Speed)** climbs/descents, or manual flight: Linearly

modulates engine thrust to maintain the selected target speed (KTS or Mach) without ever exceeding the continuous MCT threshold.

- M CRS **(Max Cruise —** 0.530 Mach**):**
  - Automatically engages Mach **mode**, enables FMS, Auto T, and MCT, and locks the cruise target to 0.530 Mach (**~311 KTAS** between FL280 and FL310, fuel flow ~65 gal/h, typical range **950 NM**).
- E CRS **(Economy Cruise —** 0.489 Mach**):**
  - Automatically engages Mach **mode**, enables FMS, Auto T, and MCT, and locks the cruise target to 0.489 Mach (**~280 KTAS** at FL310, fuel flow ~44 gal/h, maximum range up to **1,275 NM**).

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2. Standard Operating Procedures (SOP) by Flight Phase

Phase 1: Pre-Flight, Ground Handling & Taxi

1. **Engine Start & Ground Power (GPU)**:
   - On the **SYSTEMS** page of the center GTC, tap the dedicated GPU button to connect external ground power (green indicator bar illuminates). All avionics, environmental control, and cabin lights are powered at 28V without draining the main aircraft batteries.
   - *Safety Feature*: The GPU will automatically disconnect as soon as ground roll begins (groundspeed > 3 kts) or upon takeoff.
2. **Taxi**:
   - On the **SYSTEMS** page, confirm that NWS **(Nose Wheel Steering)** is active (**Green bar**) for smooth rudder-pedal directional control on the ground.
   - Extend **Flaps to 50%** ($V_{FE} = 190\text{ KIAS}$).
   - Set the initial cleared altitude using the ALT selector knob on the autopilot panel (calibrated to 100 ft per click).
3. **Runway Line-Up**:
   - Once cleared for takeoff, tap TOGA on the touchscreen:
     - Both TOGA and Auto T illuminate green.
     - The airspeed target bug sets automatically to 160 KIAS.
     - The Williams FJ33-5A engine spools up smoothly to **100% Takeoff Thrust**.

Phase 2: Takeoff & Initial Climb

1. **Rotation & Initial Climb**:
   - Smoothly rotate at 90 KIAS toward the 8°–10° pitch-up bars on the Primary Flight Display (PFD).
   - With a positive rate of climb established, retract the landing gear (**Gear UP**, limit $V_{LO} = 210\text{ KIAS}$).
   - Passing 115 KIAS, retract the **Flaps to 0% (UP)**.
2. **Transition to** MCT **and Enroute Climb (**FLC **or** VS**)**:
   - After flap retraction (and within 5 minutes of takeoff), tap MCT on the touchscreen:
     - TOGA extinguishes; MCT and Auto T remain active in green.
   - **Recommended Climb Technique —** FLC **(Flight Level Change)**:
     1. Press FLC and AP on the G1000 autopilot control panel.
     2. With speed set to 160 KIAS (up to ~FL200–FL220) and transitioning to Mach (0.400 – 0.430 M) climbing through FL250 towards FL280 – FL310, the autothrottle holds engine power fixed on the MCT **detent** while autopilot pitch control locks climb speed dead-center without oscillation.

- **Alternative Climb Technique —** VS **(Vertical Speed)**:
    1. Press VS and dial the desired vertical speed (e.g., `+1,500 ft/min` at low altitude, decaying to `+600 / +800 ft/min` above FL250).
    2. In VS mode, the Autothrottle (`Auto T` + `MCT`) modulates the thrust lever continuously to maintain the dialed airspeed up to the continuous `MCT` limit.

Phase 3: Level-Off & Cruise (`FL280 – FL310`)

Upon reaching the target altitude (e.g., `31,000 ft` / `FL310`), the autopilot smoothly captures and transitions from `FLC`/`VS` to `ALT` **(Altitude Hold)**. Simultaneously, the autothrottle transitions automatically to **active speed hold**.

Select one of **3 Cruise Profiles**:

1. **Maximum Cruise Profile (`M CRS` — `0.530 Mach` / `311 KTAS`)**:
    - Tap `M CRS` on the touchscreen.
    - The status indicators `M CRS`, `FMS`, `Mach`, `MCT`, and `Auto T` illuminate in green.
    - The target display shows `0.530 M`. The autothrottle stabilizes between `0.528 – 0.530 Mach` (`311 KTAS` ~~at ISA, fuel burn~~ `65 gal/h`, practical range **~950 NM**).
2. **Economy / Long-Range Cruise Profile (`E CRS` — `0.489 Mach` / ~`280 KTAS`)**:
    - Tap `E CRS` on the touchscreen.
    - The status indicators `E CRS`, `FMS`, `Mach`, `MCT`, and `Auto T` illuminate in green.
    - The target display shows `0.489 M`. If previously cruising faster (e.g., at `0.528 M`), the autothrottle **smoothly retards the throttle** (to 68–72% ~~TLA), allows the airframe to decelerate cleanly to~~ `0.489 Mach`~~, and then restabilizes thrust to hold~~ `0.489 Mach` ~~(fuel burn~~ `44 gal/h`, maximum range up to **1,275 NM**).
3. **Manual Cruise Speed Selection (`Mach` or `KTS`)**:
    - Tap `Mach` (or `KTS`) and turn the `SPEED SET` knob to dial your custom target (e.g., `0.510 M` or `210 KIAS`).

Phase 4: Descent

1. Set the cleared lower altitude using the autopilot `ALT` knob (e.g., `10,000 ft` or pattern altitude).
2. **Descent in** `VS` **(Vertical Speed — Recommended)**:
    - Press `VS` and set `-1,000` to `-1,500 ft/min`.
    - Switch to `KTS` mode (when descending below FL240) and set the desired descent speed using `SPEED SET` (e.g., `220 – 240 KIAS`, staying below $V_{MO} = 250\text{ KIAS}$).
    - The Autothrottle (`Auto T` + `MCT`) smoothly reduces thrust to hold the target airspeed down the descent slope.
3. **Descent in** `FLC`:
    - If `FLC` is selected with a lower altitude, the autothrottle gently brings the engine to flight idle (12% TLA) while the autopilot pitches the nose down to track the dialed airspeed. Once level at the new altitude (`ALT` capture), the autothrottle automatically spools up to restore and hold the target speed.

Phase 5: Approach, Landing & Go-Around

1. **Deceleration & Aircraft Configuration**:
    - On initial approach, dial `SPEED SET` (in `KTS` mode) to `180 KIAS`: the autothrottle reduces thrust and stabilizes at `180 KIAS`.
    - Below `210 KIAS` (V_{LO}), extend landing gear (**Gear DOWN**).

- Below 190 KIAS ($V_{FE\ 50%}$), extend **Flaps 50%** and dial 135 – 140 KIAS on the speed bug.
- On final approach, below 150 KIAS ($V_{FE\ 100%}$), extend **Flaps 100%** and reduce the speed selector to final approach reference speed (85 – 90 KIAS; stall speed in landing configuration $V_{SO} = 67\text{ KIAS}$).

2. **Landing (Autothrottle Disengagement)**:
   - At approximately 200 – 300 ft AGL (or prior to the landing flare), tap the Auto T button to **disengage the Autothrottle** (all Auto T and MCT annunciations extinguish). Manually retard your hardware throttle lever to **Flight Idle** for touchdown.
3. **Go-Around (Missed Approach)**:
   - If a missed approach is required, simply press TOGA:
     - The autothrottle advances smoothly to **100% Takeoff Thrust**.
     - Flight Director commands go-around pitch attitude (+8°) and sets target speed to 160 KIAS.
     - Immediately retract flaps to **50%**.
     - With positive climb rate, retract the landing gear (**Gear UP**).
     - Above 115 KIAS, retract flaps to **0% (UP)** and tap MCT on the GTC to resume standard climb.

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3. Cockpit Amenities & Pop-ups

- **G1000 PFD & MFD 2D Pop-ups**:
  - Clicking anywhere on the active glass screen of either the pilot PFD or copilot MFD instantly pops up a resizable, high-resolution 2D window (sim/GPS/g1000n1_popup and sim/GPS/g1000n3_popup). Clicking the window again closes it.
- **GTC Overlap-Free Layout**:
  - All secondary pages (Auto Throttle, Weight & Balance, and Systems) have their **Back** and **Home** navigation buttons placed on the dedicated right margin (POS 460.0), preventing any overlap with text boxes or system toggles.

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4. SimBrief Flight Planning Airframe

A dedicated, pre-calibrated SimBrief airframe profile matching the native XP12 Cirrus Vision Jet SF50 G2 specifications (FL310 ceiling, Williams FJ33-5A fuel flow, MTOW 6,000 lbs, accurate weights and PBN avionics) is available for 1-click import into your SimBrief fleet:

- 🌐 **Click here to import the official SimBrief Airframe Profile

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5. Electronic Flight Bag (EFB), AviTab & Live Dispatch Manager

3D Cockpit Tablet Mount & Quick-Release Latch

- The SF50 G2 includes a dedicated 3D tablet (EFB) mounted on the captain side window articulated arm (xPad_arm.obj / tablet.obj).
- **Mount / Stow Interaction**: Click the physical quick-release latch located on the top center of the tablet frame (or click anywhere on the window mount arm ball clamp, or trigger command laminar/sf50/tablet/toggle).
- **Night Illuminator Indicator**: In complete cockpit darkness, the top release latch features an integrated ambient-lit green LED illuminator, ensuring effortless tablet deployment or stowing during night flights.

Dual Mode Architecture & Home Button

- The EFB functions in two seamlessly integrated modes: **Native EFB Dispatch** and **AviTab Navigation**

Suite.

- **Physical & Screen Home Button**: A dedicated Home icon button is situated on the right-hand bezel of the tablet (and replicated at the bottom of the touchscreen). Clicking either button instantly toggles between:
    1. **Native EFB Dispatch Page**: In-cockpit fueling, passenger cabin seating, weight and balance calculations, and SimBrief synchronization.
    2. **AviTab Navigation Suite**: Navigraph digital charts, geo-referenced approach plates, interactive airport moving maps, and full SimBrief OFP viewer (requires the free AviTab plugin).

Native EFB Dispatch Page Features

- **Calibrated High-Legibility Dark UI**: Engineered specifically for high contrast, minimal cockpit glare, and crisp readability across VR headsets and 4K displays.
- **Fuel Management**:
    - One-click quick presets: `MIN` (184 kg / 405 lbs), `TABS` (429 kg / 945 lbs), `FULL` (907 kg / 2,000 lbs).
    - Fine increment buttons: `[-] 25 KG` and `[+] 25 KG` (strictly clamped within 184–907 kg).
- **Cabin Payload Management**:
    - Interactive 5-seat passenger selector with dynamic cabin silhouette indicators (`[-] PAX`, `[+] PAX`, 1 to 5 occupants).
    - Dedicated cargo/baggage weight controls (`[-] 10 KG`, `[+] 10 KG`).
- **Load Sheet & Balance**:
    - Real-time digital readout of Operating Empty Weight (`OEW: 1,692 KG`), total `PAYLOAD`, current `FUEL`, and Estimated Takeoff Weight (`EST TOW`).
- **Dynamic Weight Envelope Annunciator**:
    - `WEIGHT ENVELOPE: NORMAL (OK)` (green indicator) when gross weight is within structural limits.
    - `WEIGHT ENVELOPE: OVERWEIGHT (CAUTION)` (amber/red indicator) if total weight exceeds Maximum Takeoff Weight (**MTOW: 2,722 kg / 6,000 lbs**).

SimBrief 1-Click Synchronization & FSEconomy Reverse Payload Engine

- **User Configuration (`simbrief_user.txt`)**:
    - Open `simbrief_user.txt` in the root aircraft folder and enter your **SimBrief Pilot ID** (e.g. 123456) or your SimBrief username.
- **1-Click Live Sync**:
    - Tap `SYNC SIMBRIEF OFP` on the tablet screen (or run command `laminar/sf50/dispatch/load_simbrief`).
    - The custom XLua engine securely queries the SimBrief live API, downloads your latest dispatch OFP, and injects block fuel and payload directly into X-Plane's physics flight model in real time.
- **FSEconomy / Virtual Airline Reverse Calculation**:
    - FSEconomy and standard commercial operations mandate passenger weight at exactly **77 kg (170 lbs)** and validate total payload down to the exact single kilogram.
    - Our dispatch engine extracts SimBrief's exact total payload and automatically balances cargo using a reverse deduction formula: $$\text{Cargo (KG)} = \text{Total SimBrief Payload} - (\text{Passenger Count} \times 77.0\text{ KG})$$
    - This guarantees that your fuel, payload, and gross weight in X-Plane match your SimBrief flight plan and FSEconomy flight validation to the exact single digit!

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6. Complete Command Reference Guide

All custom aircraft commands are accessible for mapping to your physical joystick, yoke, throttle quadrant, or

keyboard shortcuts via the standard X-Plane 12 Settings menu:

System / Subsystem	Command Identifier	Description / Function
Autothrottle	`laminar/sf50/g2/autothrottle_toggle`	Toggle Master Autothrottle (`Auto T ON / OFF`)
FADEC Thrust	`laminar/sf50/g2/toga_thrust`	Command Takeoff / Go-Around 100% thrust (`TOGA`)
FADEC Thrust	`laminar/sf50/g2/mct_thrust`	Command Max Continuous Thrust limit (`MCT`)
Speed Selection	`laminar/sf50/g2/speed_mode_toggle`	Toggle speed mode between Knots (`KTS`) and `Mach`
Speed Selection	`laminar/sf50/g2/speed_mode_kts`	Directly select Knots (`KTS`) mode
Speed Selection	`laminar/sf50/g2/speed_mode_mach`	Directly select `Mach` mode
Cruise Presets	`laminar/sf50/g2/max_cruise_mach`	Activate Max Cruise profile (`0.530 M`)
Cruise Presets	`laminar/sf50/g2/econ_cruise_mach`	Activate Economy Cruise profile (`0.489 M`)
EFB Tablet	`laminar/sf50/tablet/toggle`	Mount / Stow the 3D cockpit EFB tablet
EFB Tablet	`laminar/sf50/tablet/switch_mode`	Switch EFB display between Dispatch and AviTab
Live Dispatch	`laminar/sf50/dispatch/load_simbrief`	1-Click live SimBrief OFP flight plan synchronization
Live Dispatch	`laminar/sf50/dispatch/pax_add`	Add 1 passenger to cabin payload
Live Dispatch	`laminar/sf50/dispatch/pax_sub`	Remove 1 passenger from cabin payload
Live Dispatch	`laminar/sf50/dispatch/cargo_add`	Add 10 kg cargo/baggage weight
Live Dispatch	`laminar/sf50/dispatch/cargo_sub`	Remove 10 kg cargo/baggage weight
Live Dispatch	`laminar/sf50/dispatch/fuel_min`	Load Minimum Fuel preset (`184` kg)
Live Dispatch	`laminar/sf50/dispatch/fuel_tabs`	Load Standard Tabs Fuel preset (`429` kg)
Live Dispatch	`laminar/sf50/dispatch/fuel_full`	Load Full Tanks Fuel preset (`907` kg)
Live Dispatch	`laminar/sf50/dispatch/fuel_add25`	Add 25 kg fuel to wing tanks
Live Dispatch	`laminar/sf50/dispatch/fuel_sub25`	Remove 25 kg fuel from wing tanks
Avionics & Cockpit	`sim/GPS/g1000n1_popup`	Toggle 2D resizable window for Pilot PFD
Avionics & Cockpit	`sim/GPS/g1000n3_popup`	Toggle 2D resizable window for Copilot MFD
Systems	`sim/electrical/GPU_on` / `sim/electrical/GPU_off`	Connect / Disconnect 28V Ground Power Unit (GPU)
Flight Controls	`laminar/sf50/ai/fltControl_quick_start`	Initialize flight controls and steering systems